# MILITARY SPECIFICATION

# GUIDED MISSILE, TRAINING

# ATM-78D-3, TYPE III

## 1. SCOPE

1.1 Scope. This specification establishes the design and manufacture requirements for the ATM-78D-3 Training Guided Missile Type III, referred to hereinafter as the missile.

## 2. APPLICABLE DOCUMENTS

2.1 Government documents. The following documents of the issue in effect on the date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein.

SPECIFICATIONS

Federal

| | |
|---|---|
| J-W-1177 | Wire, Magnet, Electrical |
| QQ-C-320 | Chromium Plating (Electrodeposited) |
| QQ-N-290 | Nickel Plating (Electrodeposited) |
| QQ-P-35 | Passivation Treatments for Corrosion-Resisting Steel |
| QQ-P-416 | Plating, Cadmium (Electrodeposited) |
| QQ-S-365 | Silver Plating, Electrodeposited; General Requirements for |
| TT-P-1757 | Primer Coating, Zinc Chromate, Low-Moisture-Sensitivity |

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: (Commanding Officer, Naval Air Engineering Center, Engineering Specifications and Standards Dept. (ESSD) Code 93, Lakehurst, N. J. 08733) by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.



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Military

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| MIL-M-3171 | Magnesium Alloy, Processes for Pretreatment and Prevention of Corrosion on |
| MIL-C-5541 | Chemical Conversion Coatings on Aluminum and Aluminum Alloys |
| MIL-F-7179 | Finishes and Coatings: Protection of Aerospace Weapons Systems, Structures and Parts; General Specification for |
| MIL-S-7742 | Screw Threads, Standard, Optimum Selected Series, General Specification for |
| MIL-B-7883 | Brazing of Steels, Copper, Copper Alloys, Nickel Alloys, Aluminum and Aluminum Alloys |
| MIL-S-8516 | Sealing Compound, Polysulfide Rubber, Electrical Connectors and Electric Systems, Chemically Cured |
| MIL-A-8625 | Anodic Coatings, for Aluminum and Aluminum Alloys |
| MIL-T-10727 | Tin Plating, Electrodeposited or Hot Dipped, for Ferrous and Non-Ferrous Metals |
| MIL-C-14550 | Copper Plating (Electrodeposited) |
| MIL-P-16232 | Phosphate Coatings, Heavy, Manganese or Zinc Base (for Ferrous Metals) |
| MIL-S-23586 | Sealing Compound, Electrical, Silicone Rubber, Accelerator Required |



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| MIL-M-24041 | Molding and Potting Compound, Chemically Cured, Polyurethane (Polyether-Based) |
| MIL-C-26074 | Coatings, Electroless Nickel, Requirements for |
| MIL-G-45204 | Gold Plating, Electrodeposited |
| MIL-S-45743 | Soldering, Manual Type, High Reliability Electrical and Electronic Equipment |
| MIL-R-46085 | Rhodium Plating, Electrodeposited |
| MIL-S-46844 | Solder Bath Soldering of Printed Wiring Assemblies |
| MIL-I-81550 | Insulating Compound, Electrical, Embedding, Reversion Resistand Silicone |

Naval Air Systems Command
(Code Ident. 30003)

| | |
|---|---|
| AS 1601 | Specification for Guidance Section, Missile AN/DSQ-27 |
| AS 1804 | Critical Item Product Function Specification for Guidance Section, Guided Missile AN/DSQ-36 |
| AS 1806 | Critical Item Product Function Specification for Guidance Section Guided Missile AN/DSQ-36A |

STANDARDS

Military

| | |
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| MIL-STD-454 | Standard General Requirements for Electronic Equipment |



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| MIL-STD-709 | Ammunition Color Coding |
| MIL-STD-1303 | Painting of Naval Ordnance Equipment |
| MIL-STD-1523 | Age Controls of Age-Sensitive Elastomeric Materiel |
| MIL-STD-45662 | Calibration System Requirements |

DRAWINGS

Naval Air Systems Command 30003

| | |
|---|---|
| 423-300050 | Guidance Section, Guided Missile |
| 423-510120 | Steering Control Section, Dummy |
| 911AS900 | Guided Missile, Training, ATM-78D-3 |
| 911AS901 | Ordnance Section Assembly |
| 911AS903 | Autopilot-Battery Section Assembly |
| 911AS910 | Block Diagram, Guided Missile, Training, ATM-78D-3, Type III |
| 911AS911 | Guided Missile, Training, ATM-78D-3, Type III |
| 911AS1000 | Guidance Section, Guided Missile |
| 911AS1020 | Guidance Section, Guided Missile |
| 1181AS200 | Rocket Motor, Mark 69 Mod 2, Inert |
| 2456003 | Spacer, Rocket Motor, Machined |
| 2622700 | Control Surface Assembly |
| 2639027 | Control Surface Assembly |

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specified procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)



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2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated the issue in effect on date of invitation for bids or request for proposal shall apply.

American Society for Testing and Materials

| | |
|---|---|
| ASTM D2863 | Standard Method of Test for Flammability of Plastics using the Oxygen Index Method |

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race St., Philadelphia, PA 19103.)

Naval Air Systems Command

| | |
|---|---|
| NAVAIR 01-5GMAA-20.1 | Technical Manual, Assembly and Checkout Instructions with Illustrated Parts Breakdown (Intermediate Level), Standard Arm Missile AGM-78D-2 and ATM-78D-2 |

(Application for copies should be addressed to Naval Air Technical Services Facility, 700 Robbins Ave., Philadelphia, PA 19111.)

3. REQUIREMENTS

3.1 General design. The missile shall be in accordance with this specification and Drawing 911AS900.

3.1.1 Physical compatibility. The missile shall be physically compatible with the Standard ARM AGM-78D-1 missile.

3.1.2 Electrical compatibility. The missile shall provide missile electrical functions as specified on Drawing 911AS911.

3.2 Characteristics.

3.2.1 Performance. The missile shall comply with the performance requirements specified for the guidance section in AS 1601, AS 1804, or AS 1806, as applicable.

3.2.2 Physical characteristics.

3.2.2.1 Dimensional limitations. The missile size and shape shall be as specified on Drawing 911AS900.

3.2.2.2 Weight. The missile weight shall be 1355.5 ±25.0 pounds.



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3.2.2.3 Center of gravity (cg). The missile cg shall be at station 105.28 ±1.00.

3.3 Design and construction.

3.3.1 Major sections. The missile shall be composed of the major sections specified in 3.3.1.1 through 3.3.1.6 (see Drawing 911AS910).

3.3.1.1 Guidance section. The guidance section shall be as specified on Drawing 423-300050, 911AS1000, or 911AS1020. The preferred guidance section shall be P/N 423-300050. Guidance section P/N's 911AS1000 and 911AS1020 shall be alternate and interchangeable.

3.3.1.2 Ordnance section. The ordnance section shall be as specified on Drawing 911AS901, and shall consist of a machined shell, wiring harness, and dummy fuze. Four identical dummy antennas shall be painted on the section.

3.3.1.3 Autopilot/battery section. The autopilot/battery (AP/B) section shall be as specified on Drawing 911AS903. All guidance section TLM functions that are normally telemetered shall be available at an all-environmental dorsal connector. The AP/B section shall include a time/event indicator assembly to provide the capability for training missile handling personnel in recording time and event information.

3.3.1.4 Rocket motor. The rocket motor shall be an inert MK 69 Mod 2 rocket motor (see Drawing 1181AS200).

3.3.1.5 Rocket motor spacer. The rocket motor spacer section shall be as specified on Drawing 2456003.

3.3.1.6 Steering control section. The steering control section shall be a ballasted unit (see Drawing 423-510120), with locked control surfaces [see Drawing 2622700 or 2639027 (preferred)].

3.3.2 Materials, processes, and parts.

3.3.2.1 Materials.

3.3.2.1.1 Toxic materials. Unless otherwise specified in the contract, materials which produce harmful toxic effects under conditions indicated in this specification shall not be used.

3.3.2.1.2 Flammable materials. Unless otherwise required for the design, materials which are not self-extinguishing, as determined by ASTM D2863, or which are capable of causing an explosion shall not be used.

3.3.2.1.3 Magnesium. Magnesium shall not be used unless otherwise specified in the contract. If the use of magnesium is required, it shall be treated as specified in MIL-M-3171.

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3.3.2.1.4 Elastomeric materials. Elastomeric materials shall comply with the age controls of MIL-STD-1523. Elastomeric potting and encapsulating compounds shall comply with MIL-S-8516, MIL-S-23586, MIL-M-24041, or MIL-I-81550.

3.3.2.1.5 Moisture and fungus resistance. Materials resistant to moisture and fungus, as defined in MIL-STD-454, requirement 4, shall be used whenever practicable. If fungus nutrient materials are used, they shall meet the requirements of MIL-STD-454, requirement 4, group II. Fungus nutrient materials may be used in permanently hermetically sealed applications.

3.3.2.1.6 Plastics. Plastic laminates containing cotton-fabric base or plastic molding compounds containing cellulose products shall not be used (see 3.3.2.1.5). Plastic materials shall be completely cured prior to installation to preclude outgassing during storage.

3.3.2.1.7 Wiring. Hookup wire shall be in accordance with MIL-STD-454, requirement 20. Printed wiring for electronic equipment shall be in accordance with MIL-STD-454, requirement 17. Magnet wire shall be in accordance with J-W-1177.

3.3.2.2 Processes.

3.3.2.2.1 Protective coatings and surface treatments. Unless otherwise specified herein, finishes of machined, cast, forged, and welded surfaces shall be the minimum required for proper operation, good appearance, and economy of manufacture. Protective coatings shall not be used in areas which will lose that coating during the normal course of operation, testing, and maintenance. Screws, bolts, slip-fits, screw-threaded inserts, and press-fits shall be assembled using wet zinc chromate primer in accordance with TT-P-1757.

3.3.2.2.2 Finishes. Finishes of metal surfaces shall be as specified in MIL-F-7179 and on the applicable drawings. Painting shall be in accordance with MIL-STD-1303. Color coding shall be in accordance with MIL-STD-709.

3.3.2.2.3 Plating. Unless otherwise specified in the contract, plating of metal surfaces shall be in accordance with the requirements of the applicable specification for each of the following types of plating:

| | | |
|---|---|---|
| a. | Chromium: | QQ-C-320, class 2 |
| b. | Nickel: | QQ-N-290, class 2, or MIL-C-26074, class 1 or 2 |
| c. | Cadmium: | QQ-P-416, class 2, type II |
| d. | Silver: | QQ-S-365, type II, grade B |
| e. | Gold: | MIL-G-45204, type I, grade C or D |
| f. | Rhodium: | MIL-R-46085, type I, class 2 |
| g. | Copper: | MIL-C-14550 |
| h. | Tin | MIL-T-10727 |



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Steel parts not in moving contact, other than corrosion resistant steel, shall be cadmium plated in accordance with QQ-P-416. Sliding or rotating parts shall be chromium plated in accordance with QQ-C-320, or treated in a manner to prevent corrosion. Cadmium plating shall not be used on parts in moving contact, or on parts in contact with titanium.

3.3.2.2.4 Zinc plating. Zinc plating shall not be used, due to its low resistance to corrosion.

3.3.2.2.5 Phosphate coating. Phosphate coatings for unpainted ferrous metal surfaces shall be in accordance with MIL-P-16232.

3.3.2.2.6 Aluminum surfaces. Parts fabricated from aluminum or aluminum alloys shall be anodized in accordance with MIL-A-8625 or chemically coated in accordance with MIL-C-5541. Painting of aluminum surfaces following either of the above specified treatments is permissible.

3.3.2.2.7 Corrosion of metal parts. Dissimilar metals shall not be used in direct contact unless suitably protected to prevent electrolytic corrosion. Selection and protection of dissimilar metal combinations shall be in accordance with MIL-STD-454, requirement 16 for class 3 equipment. Corrosion-resistance requirements for ferrous alloys used in electronic equipment shall be in accordance with MIL-STD-454, requirement 15.

3.3.2.2.8 Soldering. Soldering processes shall be in accordance with MIL-S-45743, MIL-S-46844, or MIL-STD-454, requirement 5. Solder joints shall not depend solely on the solder for mechanical strength, except for joints on relatively light parts which have been proven by actual use to have adequate strength.

3.3.2.2.9 Brazing. Brazing of metals shall be in accordance with MIL-B-7883.

3.3.2.2.10 Passivation. Unless otherwise specified herein, corrosion and heat-resistant steel shall be processed and passivated in accordance with QQ-P-35.

3.3.2.3 Parts.

3.3.2.3.1 Threaded parts. Threading of parts shall be in accordance with MIL-S-7742.



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3.3.3 Identification and marking. The missile shall be identified in accordance with Drawing 911AS900. The identification shall be stenciled in contrasting color at two places on the side of the missile, in an area plainly visible during use. Areas of soft spots on ordnance sections shall be painted with a contrasting color and stenciled to indicate the nature of the area.

3.3.4 Government-furnished property. One of the applicable following listed containers will be furnished by the Government for each missile to be delivered (see 6.4):

a. Guided Missile Container MK 372 MOD 4, P/N 2644150
b. Guided Missile Container CNU-183/E, P/N 67A 220D1

3.3.5 Workmanship. The missile, including all parts, components, and subassemblies shall be constructed and finished in a thoroughly professional manner. Particular attention shall be given to neatness and thoroughness of soldering, marking of parts, subassemblies, wiring, welding and brazing, riveting, finishes, machine-screw assemblage, freedom from burrs, sharp edges, dirt, chips, other foreign matter and visible defects which are likely to affect or render the missile unsatisfactory for the operation or function intended. Workmanship for electronic equipment shall be in accordance with MIL-STD-454, requirement 9.

## 4. QUALITY ASSURANCE PROVISIONS

4.1 General. This section establishes the total requirements for verifying missile performance, operability, and physical characteristics. The quality assurance provisions shall consist of quality conformance inspection (see 4.2).

4.1.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the contractor may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1.1 Procuring activity acceptance. The inspections required herein are those by which the contractor shall provide the basis for acceptance of the missile round.



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4.1.1.2 Inspection equipment, facilities, and conditions. Inspection equipment and facilities shall be of adequate quality and quantity to permit performance of the required examinations and tests. Unless otherwise specified in the contract, all examinations and test shall be performed within the following ranges of ambient temperature, humidity, and barometric pressure:

| | | |
|---|---|---|
| a. | Temperature: | +22° ±2°C (+72° ±4°F) |
| b. | Relative humidity: | 20 to 95 percent |
| c. | Barometric pressure: | Sea level to 7000 feet |

4.1.1.2.1 Test equipment error. Equipment used to measure parameters shall introduce no overall errors greater than 10 percent of tolerance on the parameter, or, in the case of a single limit tolerance, the test equipment error shall be applied to the specified parameter in such a manner as to assure that the specified parameter is being met. Test equipment calibration shall be in accordance with MIL-STD-45662.

4.2 Quality conformance inspection.

4.2.1 Physical, visual, and analytical inspections.

4.2.1.1 Weight and cg. The weight and cg shall be verified for compliance with 3.2.2.2 and 3.2.2.3.

4.2.1.2 Materials, processes, and parts. It shall be ascertained by the inspection of data that all materials, processes, and parts have been inspected and found to comply with their respective specifications and applicable drawings, and that all specified manufacturing processes have been followed during production.

4.2.1.3 Identification and workmanship inspection. The missile shall be examined to determine compliance with the identification and marking requirements of 3.3.3, and the workmanship requirements of 3.3.5.

4.2.2 Performance inspection methods. The functional performance of the missile shall be tested in accordance with AS 1601, AS 1804, or AS 1806, as applicable.

## 5. PACKAGING

5.1 Preservation, packing, and packaging. Unless otherwise specified in the contract or purchase order, preservation, packaging, and packing shall be in accordance with NAVAIR 01-5GMAA-20.1, and missiles shall be packaged in Government furnished containers (see 3.3.4 and 6.4).



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6. NOTES

6.1 Intended use. The missile will be used to simulate prelaunch conditions for ground and flight training purposes.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of issue in effect, both of this specification and the applicable drawing (see 2.1).

b. Applicable level of preservation, packaging, and packing (see 5.1).

c. Use of magnesium, if required (see 3.3.2.1.3)

d. Permission on use of toxic materials (see 3.3.2.1.1)

e. Plating, if other than specified in 3.3.2.2.3

f. Responsibility for inspections, if other than specified in 4.1.1.

6.2.1 Oral statements. No oral statement by any person or persons should be allowed in any manner or degree to modify or otherwise affect the requirements of any part of this specification or any specification, standard, publication, drawing, or document required herein.

6.3 Definition. The following definition applies to the specification when the term defined is used herein.

6.3.1 Continuous production. Continuous production consists of a designated production line producing the items by means of the same processes, material requirements, controls and design. Any recurring shutdown, such as weekends, holidays or preventive maintenance of the line should not affect the status of continuous production. Interruption in production exceeding 30 calendar days, regardless of cause, terminates continuous production.

6.4 Government-furnished property. The contracting officer should arrange to furnish the property listed in 3.3.4.

Preparing activity:
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DOCUMENT IDENTIFIER *(Number)* AND TITLE MIL-D-85435(AS)
Military Specification Guided Missile, Training ATM-78D-3, Type III

NAME OF ORGANIZATION AND ADDRESS OF SUBMITTER

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